

Lecture 6: Analysis

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- ▶ A central concept in analysis is “convergence”
- ▶ Building on convergence, we have continuity, completeness, compactness, etc
- ▶ Important theorems then follow

Topics to be covered today

- ▶ Metric Spaces
- ▶ Sequences and Convergence
- ▶ Completeness
- ▶ Compactness
- ▶ Continuity

- 1 Metric Spaces
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- ▶ What is the distance between two real numbers a and b ?
- ▶ What is the distance between two points on the plane?
- ▶ What is the distance between $f(x) = 0$ and $f(x) = \sin(x)$?

Definition 1

A **distance** or **metric** on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ such that for all $x, y, z \in X$ the following holds:

- ① $d(x, y) \geq 0$ and $d(x, y) = 0$ iff $x = y$
- ② $d(x, y) = d(y, x)$
- ③ triangle inequality: $d(x, y) \leq d(x, z) + d(z, y)$

A **metric space** (X, d) is a set X together with a distance function d on X .

- ▶ On $\mathbb{R}$, $d(x, y) = |x - y|$ is a metric (prove it)
- ▶ More generally, the **Euclidean** metric on $\mathbb{R}^n$:

$$d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$

- ▶ The metric induced by the **sup norm** on the set of bounded real-valued functions on X :

$$d(f, g) = \|f - g\|_{\infty} = \sup_{x \in X} |f(x) - g(x)|$$

Definition 2

- ▶ An **open ball** with center a and radius r is defined by

$$B_r(a) := \{x \in X : d(x, a) < r\}$$

- ▶ A subset Y of X is a **neighborhood** of $a \in X$ if there exists $r > 0$ such that $B_r(a) \subset Y$
- ▶ A subset S of X is **bounded** if $S \subset B_r(a)$ for some $a \in X$ and $r > 0$

Open Balls and Neighborhoods

Examples

- ▶ $(0, 2)$ is an open ball $B_1(1)$
- ▶ Any open interval (a, b) in $\mathbb{R}$ is an open ball (prove it)

Proposition 3

If S is a bounded subset of a metric space X , then for every $y \in X$ there exists $\rho > 0$ such that $S \subset B_\rho(y)$.

Prove it in the assignment

Definition 4

Let S be a subset of a metric space (X, d) .

- ▶ A point $x \in X$ is an **interior** (resp. **exterior**) point of S if some open ball centered at x is a subset of S (resp. S^c).
- ▶ If every open ball centered at x contains at least one point in S and at least one point in S^c , then x is a **boundary** point of S .
- ▶ The set of all interior (exterior) points of S is called the interior (exterior) of S and is denoted by $\text{int } S$ ($\text{ext } S$).
- ▶ The set of all boundary points of S is called the boundary of S and is denoted by ∂S .

Interior, Exterior, Boundary, and Closure

Examples

Let $X = \mathbb{R}$ and $S = [0, 1)$

- ▶ $x = 1$, $x = 0$, $x = 0.5$, $x = 2$?
- ▶ What is $\text{int } S$? What is ∂S ?

Let $X = [0, 1]$ and $S = [0, 1)$

- ▶ $x = 1$, $x = 0$, $x = 0.5$?
- ▶ What is $\text{int } S$? What is ∂S ?

Definition 5

- ▶ A point $x \in X$ is a **limit point** of S if every open ball centered at x contains at least one element of S other than x .
- ▶ A point $x \in S$ is an **isolated point** of S if some open ball centered at x contains no element of S other than x .
- ▶ The **closure** of S is the union of S and the set of limit points of S , and is denoted by $\bar{S}$.

Proposition 6

- 1 If x is a limit point of S , then every open ball centered at x contains an infinite number of points from S
- 2 $\bar{S} = (\text{ext } S)^c$
- 3 $\bar{S} = \text{int } S \cup \partial S$

Prove some using the definitions

Interior, Exterior, Boundary, and Closure

Examples

- ▶ If $S = \{1/n : n \in \mathbb{N}\} \subset \mathbb{R}$, then every point in S is an isolated point. The only limit point is 0.
- ▶ Consider $\mathbb{Q} \subset \mathbb{R}$. Then $\text{int } \mathbb{Q} = \emptyset$, $\partial \mathbb{Q} = \mathbb{R}$, and $\bar{\mathbb{Q}} = \mathbb{R}$.

A set is open if all its elements are interior points.

Definition 7

- ▶ A set $S \subset X$ is **open** iff $S \subset \text{int } S$.
- ▶ A set $S \subset X$ is **closed** iff S^c is open.

Proposition 8

- 1 Any open ball is an open set
- 2 A set S is closed iff $S = \bar{S}$
- 3 The sets $\bar{S}$ and ∂S are closed

Theorem 9

- ① *Any union of open sets is open; any intersection of closed sets is closed.*
- ② *Any finite intersection of open sets is open; any finite union of closed sets is closed.*

Example: $(0, 1) = \bigcup_{n=2}^{\infty} [1/n, 1 - 1/n]$ (prove this)

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Definition 10

- ▶ A **sequence** in a set X is a function from $\mathbb{N}$ to X . We write $x_1, x_2, \dots$, or $(x_n)_{n=1}^{\infty}$, or simply (x_n) for the sequence.
- ▶ Given a sequence (x_n) , a **subsequence** is a sequence (x_{n_i}) where (n_i) is a strictly increasing sequence in $\mathbb{N}$.

Examples: $x_n = 1/n$, $x_n = 2n + 1$, etc.

Definition 11

A sequence (x_n) in a metric space (X, d) **converges** to a **limit** $x \in X$, written as $x_n \rightarrow x$ or $\lim_{n \rightarrow \infty} x_n = x$, if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow d(x_n, x) < \epsilon.$$

Proposition 12

A sequence in a metric space can have at most one limit.

Proof: Suppose $x_n \rightarrow x$ and $x_n \rightarrow y$ as $n \rightarrow \infty$. Suppose $x \neq y$ and let $d(x, y) = r > 0$.

There exists $N_1, N_2 \in \mathbb{N}$ such that

$$n \geq N_1 \Rightarrow d(x_n, x) < r/2$$

$$n \geq N_2 \Rightarrow d(x_n, y) < r/2$$

Let $N = \max\{N_1, N_2\}$. Then $d(x, y) \leq d(x, x_N) + d(y, x_N) < r$, which is a contradiction.

(Bounding the distance between two points using triangle inequality is a very important method that will be used over and over again.)

Definition 13

A sequence (x_n) is **bounded** if the set $\{x_n : n \in \mathbb{N}\}$ is bounded.

Proposition 14

A convergent sequence in a metric space is bounded.

Proof: Let $x_n \rightarrow x$. There exists an $N \in \mathbb{N}$ such that $d(x_n, x) < 1$ for all $n \geq N$. Let

$$M = \max\{d(x_1, x), d(x_2, x), \dots, d(x_N, x), 1\}.$$

Then $d(x_n, x) \leq M$ for all $n \in \mathbb{N}$.

(Using convergence to handle the “tail” of a sequence is an important method.)

Theorem 15

Let S be a subset of X . Then $x \in \bar{S}$ iff there is a sequence $(x_n) \subset S$ such that $x_n \rightarrow x$.

Corollary 16

Let S be a subset of X . Then S is closed iff

$$(x_n) \subset S \text{ and } x_n \rightarrow x \Rightarrow x \in S.$$

Proof: Suppose $(x_n) \subset S$ and $x_n \rightarrow x$. If $x_n = x$ for some n , then $x \in S \subset \bar{S}$. Otherwise, every open ball centered at x contains a term $x_n \in S$ other than x , so x is a limit point of S and hence $x \in \bar{S}$.

Conversely, suppose $x \in \bar{S}$. If $x \in S$, take the constant sequence $x_n = x$. If $x \notin S$, choose $x_n \in B_{1/n}(x) \cap S$ for each $n \in \mathbb{N}$. Then $d(x_n, x) < 1/n$, so $x_n \rightarrow x$.

Proposition 17

Suppose $x_n \rightarrow x$ and $y_n \rightarrow y$. Then

- ① $x_n + y_n \rightarrow x + y$
- ② $cx_n \rightarrow cx$ for all $c \in \mathbb{R}$
- ③ $x_n y_n \rightarrow xy$
- ④ $1/x_n \rightarrow 1/x$ if $x_n \neq 0$ and $x \neq 0$

Proposition 18

- ① If $0 \leq x_n \leq y_n$ for all $n \geq N$, and $y_n \rightarrow 0$, then $x_n \rightarrow 0$.
- ② If $x_n \leq y_n$ for all $n \geq N$, $x_n \rightarrow x$, and $y_n \rightarrow y$, then $x \leq y$.

Theorem 19

Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

The theorem can be proved using the Completeness Axiom.

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- ▶ The definition for convergence requires that we know the limit a priori.
- ▶ We would like a criterion for convergence that does not depend on the limit itself.

Definition 20

Let $(x_n) \subset X$. Then (x_n) is a **Cauchy sequence** if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \Rightarrow d(x_m, x_n) < \epsilon.$$

Proposition 21

- ① *If a sequence is Cauchy, then it is bounded.*
- ② *If a sequence converges, then it is Cauchy.*

Similar to convergent sequences, Cauchy sequences are also bounded, but the second result above suggests that Cauchy is weaker.

When is a Cauchy sequence convergent?

Definition 22

A metric space (X, d) is **complete** if every Cauchy sequence in X has a limit in X .

Proposition 23

$\mathbb{R}^n$ with the Euclidean metric is a complete metric space.

Proposition 24

Let (X, d) be a metric space and let $S \subset X$.

- ① If (S, d) is complete, then S is closed.
- ② If (X, d) is complete, then S is closed in X iff (S, d) is complete.

Proof: 1. To prove that S is closed, we pick any sequence $x_n \subset S$ converging to $x \in X$. Then (x_n) is Cauchy in S and thus has a limit in S .

2. Suppose S is closed. Since any Cauchy sequence in S is a Cauchy sequence in X , and thus has a limit in X . Since S is closed, any sequence $(x_n) \subset S$ that converges has a limit in S . Therefore, (S, d) is complete. The other direction follows from 1.

Definition 25

Let (X, d) be a metric space. We say $F : S \subset X \rightarrow X$ is a **contraction** if there exists $0 \leq \lambda < 1$ such that

$$d(F(x), F(y)) \leq \lambda d(x, y)$$

for all $x, y \in S$.

Definition 26

We say $x^* \in S$ is a **fixed** point of $F : S \subset X \rightarrow X$ if $F(x^*) = x^*$.

Theorem 27 (Contraction Mapping Theorem)

Let (X, d) be a complete metric space and let $F : X \rightarrow X$ be a contraction. Then F has a unique fixed point x^ and $F^n(x) \rightarrow x^*$ as $n \rightarrow \infty$ for all $x \in X$.*

Notes: here $F^n(x)$ means apply F iteratively n times:

$$x_1 = F(x), x_2 = F(x_1), x_3 = F(x_2), \dots, x_n = F(x_{n-1})$$

- ▶ In economics, the contraction mapping theorem is mainly used in dynamic programming
- ▶ The Bellman operator is a contraction mapping

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Theorem 28

If a sequence in a metric space converges, then every subsequence converges to the same limit as the original sequence.

Theorem 29

If a Cauchy sequence (x_n) in a metric space has a subsequence converging to x , then $x_n \rightarrow x$.

Proof: The first is by applying the definition. To prove the second theorem, pick an arbitrary $\epsilon > 0$. There exists $N \in \mathbb{N}$ such that $d(x_m, x_n) < \epsilon/2$ for all $m, n \geq N$. If a subsequence (x_{n_i}) converges to x , then there exists $M \in \mathbb{N}$ such that $d(x_{n_i}, x) < \epsilon/2$ for all $i \geq M$. Choose one $j \geq M$ such that $n_j \geq N$. Then for every $m \geq N$,

$$d(x, x_m) \leq d(x, x_{n_j}) + d(x_{n_j}, x_m) < \epsilon.$$

Theorem 30 (Bolzano-Weierstrass Theorem)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

The same result is not true for an arbitrary metric space!

Corollary 31

If $S \subset \mathbb{R}^n$, then S is closed and bounded iff every sequence from S has a subsequence which converges to a limit in S .

Proof: First suppose S is closed and bounded. Since S is bounded, every sequence has a convergent subsequence. Since S is closed, the limit is in S .

For the other direction, closedness is obvious. We prove S is bounded by contradiction. Suppose S is unbounded. Then for every $n \in \mathbb{N}$, there exists $x_n \in S$ such that $|x_n| > n$. Any subsequence of (x_n) is unbounded and thus cannot converge.

Definition 32

A subset S of a metric space (X, d) is **compact** if every sequence from S has a subsequence that converges to a limit in S .

This definition is actually called *sequential compactness*, but it is equivalent to compactness for metric spaces.

The Bolzano-Weierstrass Theorem implies the following theorem (although it was proved independently).

Theorem 33 (Heine-Borel Theorem)

A subset of $\mathbb{R}^n$ is compact iff it is closed and bounded.

Proposition 34

If a subset of a metric space (X, d) is compact, then it is bounded and closed (in X).

Proof: Suppose $S \subset X$ is compact. Then for any sequence $(x_n) \subset S$ that converges to $x \in X$, there exists a subsequence that converges to a limit in S . By a previous theorem, the limit must be x . Hence S is closed.

Suppose S is not bounded. Then we can build a sequence that has no convergent subsequence.

Proposition 35

Let S be a subset of a compact metric space (X, d) . Then S is compact iff S is closed in X .

Proof: If S is compact, then it is closed. Suppose that S is closed. Since X is compact, any sequence in $S \subset X$ has a convergent subsequence in X . Since S is closed, the limit is also in S . Hence S is compact.

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Definition 36

Let $f : (A \subset X) \rightarrow Y$ where X and Y are metric spaces and let a be a limit point of A . If for every sequence $(x_n) \subset A \setminus \{a\}$, $x_n \rightarrow a$ implies $f(x_n) \rightarrow b$, then we say $f(x)$ approaches b as x approaches a , or f has **limit** b at a , and write

$$f(x) \rightarrow b \text{ as } x \rightarrow a,$$

or

$$\lim_{x \rightarrow a} f(x) = b.$$

We have an equivalent definition of a limit using the usual ϵ - δ language

Theorem 37

Suppose (X, d) and (Y, ρ) are metric spaces, $A \subset X$, $f : A \rightarrow Y$, and a is a limit point of A . Then the following are equivalent

- ❶ $\lim_{x \rightarrow a} f(x) = b$
- ❷ For every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$x \in A \setminus \{a\} \text{ and } d(x, a) < \delta \Rightarrow \rho(f(x), b) < \epsilon$$

Definition 38

Let $f : A(\subset X) \rightarrow Y$ where X and Y are metric spaces and let $a \in A$. We say f is **continuous** at a if a is an isolated point of A or if a is a limit point and $\lim_{x \rightarrow a} f(x) = f(a)$.

If f is continuous at every $a \in A$, then we say f is **continuous**. The set of all such continuous functions is denoted by $\mathcal{C}(A; Y)$ or $\mathcal{C}(A)$ if $Y = \mathbb{R}$.

We also have equivalent definitions of continuity

Proposition 39

Suppose (X, d) and (Y, ρ) are metric spaces, $A \subset X$, $f : A \rightarrow Y$, and $a \in A$. Then the following are equivalent:

- ❶ *f is continuous at a*
- ❷ *whenever $(x_n) \subset A$ and $x_n \rightarrow a$, then $f(x_n) \rightarrow f(a)$*
- ❸ *for each $\epsilon > 0$, there exists $\delta > 0$ such that*

$$x \in A \text{ and } d(x, a) < \delta \Rightarrow \rho(f(x), f(a)) < \epsilon$$

Theorem 40

- 1 Let f and g be two real-valued functions. If f and g are continuous at a , then $f + g$, fg , and f/g (if $g(a) \neq 0$) are continuous at a
- 2 If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a

Theorem 41 (Intermediate Value Theorem)

Let f be a continuous real function on the interval $[a, b]$. If $f(a) < f(b)$ and c is such that $f(a) < c < f(b)$, then there exists a point $x \in (a, b)$ such that $f(x) = c$.

Theorem 42

Let $f : X \rightarrow Y$ where (X, d) and (Y, ρ) are metric spaces. Then the following are equivalent

- 1 f is continuous
- 2 $f^{-1}(E)$ is open in X whenever E is open in Y
- 3 $f^{-1}(C)$ is closed in X whenever C is closed in Y

The continuous image of an open set is not necessarily open and the continuous image of a closed set is not necessarily closed.

For example, let $f(x) = x^2$ and then $f((-1, 1)) = [0, 1)$; let $f(x) = e^x$ and $f(\mathbb{R}) = (0, \infty)$

Theorem 43

Let $f : K(\subset X) \rightarrow Y$ be a continuous function, where X and Y are metric spaces, and K is compact. Then

- ① $f(K)$ is compact
- ② If $Y \subset \mathbb{R}$ and K is nonempty, then f is bounded (above and below) and has a maximum and a minimum

Examples:

- ▶ Let $f(x) = 1/x$ for $x \in (0, 1]$
- ▶ Let $f(x) = x$ for $x \in [0, 1)$
- ▶ Let $f(x) = 1/x$ for $x \in [1, \infty)$

The Space of Bounded Continuous Functions

Definition 44

The set of *bounded* continuous functions $f : A \rightarrow Y$ is denoted by $\mathcal{BC}(A; Y)$.

Theorem 45

Suppose $A \subset X$, (X, d) is a metric space and (Y, ρ) is a complete metric space. Then $\mathcal{BC}(A; Y)$ is a complete metric space under the **uniform metric** d_u defined by

$$d_u(f, g) := \sup_{x \in A} \rho(f(x), g(x)).$$

Several important results:

- 1 $(\mathcal{BC}(A; Y), d_u)$ is a metric space
- 2 $\mathcal{BC}(A; Y)$ is closed under uniform convergence: if $f_n \rightarrow f$ in the uniform metric, then f is also bounded continuous